

Two-dimensional OLCT of angularly periodic functions in polar coordinates

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ABSTRACT

Searching for novel signal processing theories and methods has always been a research hotspot in the field of modern signal processing. The existing transform methods for dealing with non-stationary signals are based on traditional Cartesian coordinates, which are very inconvenient to deal with signals naturally described by polar coordinates. This paper studies two-dimensional offset linear canonical transform (OLCT) in polar coordinates. Firstly, the definition of the two-dimensional OLCT in polar coordinates is proposed, and the offset linear canonical Hankel transform (OLCHT) formula is derived. Secondly, the relationship between the OLCT and the OLCHT is revealed through the angular periodic function. Then, some properties of the two-dimensional OLCT in polar coordinates are proved based on the mentioned relationship. Finally, the proposed two-dimensional OLCT is applied in the computed tomography. The effectiveness and feasibility of the proposed method are verified by simulation experiments.

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1. Introduction

In recent years, the linear canonical transform (LCT) has become an extensive signal analysis tool in applied mathematics, signal processing and optical system analysis fields [1–5]. Compared with the LCT, the offset linear canonical transform (OLCT) adds two additional parameters τ and η (corresponding to time offset and frequency modulation, respectively) to its basis, and hence, it is more general than the original LCT [6,7]. The OLCT is a linear integral transform with six parameters, and a time-shifted and frequency-modulated version of the LCT. Many linear transforms are widely used in practical applications, such as the Fourier transform (FT), fractional Fourier transform (FRFT) [8–10], offset FRFT [11,12], Fresnel transform [13] and LCT [14–16], and they are all special cases of the OLCT. Therefore, studying the OLCT and developing related theories can further enrich the theoretical system of signal processing based on linear canonical theory.

As a generalized form of the FRFT and the LCT, the OLCT has been further developed, and many meaningful results have been obtained, such as convolution theorem [6,7], sampling theorem [17,18], uncertainty principles [19,20], etc. All these results are dealing with one-dimensional (1D) signals. But in practical problems, some signals are particularly well represented in polar coordinates as they are complicated in traditional Cartesian coordinates, such as photoacoustics [21], computed tomography (CT) [22], and magnetic resonance imaging [23] fields. Therefore, it is very important to explore whether the results existing in 1D case can be extended to the 2D polar coordinate case.

The 2D polar coordinate transform indicates that the general function in the Cartesian coordinate system produces a special function, which forms an angular function with a period of 2π in the polar coordinate system, and vice versa. Therefore, the aim of this article is to obtain a 2D research of the specific function class in the polar coordinate system. Firstly, a new representation of the 2D OLCT in polar coordinates is studied, and the expression of the offset linear canonical Hankel transform (OLCHT) is derived in detail. Secondly, using an extension of the exponent expansion formula [24], the relationship between the 2D OLCT and the OLCHT is obtained. Thirdly, based on this relationship, the spatial shift and convolution theorems of the 2D OLCT are studied. Finally, an application of the proposed 2D OLCT in the CT field is presented. Since nonbandlimited signals in the FT domain may be bandlimited signals in the OLCT domain, we can obtain

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new insight by using the proposed method of this paper in the CT field. The results of this paper not only study some useful properties of the 2D OLCT in polar coordinates, but also provide a theoretical basis for its application in optics and signal processing.

The paper is organized as follows. In Section 2, some basic knowledge of the 2D OLCT in the Cartesian coordinate system is provided. In Section 3, the definitions of the 2D OLCT and the OLCST in polar coordinates are given, and a relationship between the 2D OLCT and the OLCST is discussed. In Section 4, the space shift theorem and convolution theorem of the 2D OLCT are proved. In Section 5, some applications of the 2D OLCT in polar coordinates are provided. In Section 6, the article is concluded.

2. Preliminaries

In this section, we provide some necessary background and notations of the OLCT.

The 1D OLCT with real parameters of $M = (a, b, c, d, \tau, \eta)$ and a signal $f(t)$ is defined by [6,8,9,17]

$$F^M(u) = O_L^M[f(t)](u) = \begin{cases} \int_{\mathbb{R}} f(t) h_M(t, u) dt, & b \neq 0 \\ \sqrt{d} e^{i \frac{cd}{2}(u-\tau)^2 + i\eta u} f[d(u-\tau)], & b = 0 \end{cases} \quad (1)$$

where d stands for differential notation, the kernel $h_M(t, u)$ is

$$h_M(t, u) = K_M e^{i \left[\frac{a}{2b} |t|^2 + \frac{1}{b} t(\tau - u) - \frac{1}{b} u(d\tau - b\eta) + \frac{d}{2b} |u|^2 \right]}, \quad (2)$$

and $K_M = \frac{1}{\sqrt{i2\pi|b|}} e^{i \frac{d\tau^2}{2b}}$, a, b, c, d, τ, η are real numbers satisfying $ad - bc = 1$. It can be seen from Reference [8] that we generally use

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{pmatrix} t \\ u \end{pmatrix} + \begin{pmatrix} \tau \\ \eta \end{pmatrix}$$

to represent the six parameters $M = (a, b, c, d, \tau, \eta)$ of the 1D OLCT.

Let $\mathbf{t} = (t_1, t_2)$, $\mathbf{u} = (u_1, u_2)$, where $\mathbf{t}$ and $\mathbf{u}$ are 2D variables of time and OLCT domains, respectively. Using a similar idea to study the 2D OLCT, we also give the following parameter representations:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{pmatrix} \mathbf{t} \\ \mathbf{u} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\tau} \\ \boldsymbol{\eta} \end{pmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} t_1 & t_2 \\ u_1 & u_2 \end{bmatrix} + \begin{bmatrix} \tau_1 & \tau_2 \\ \eta_1 & \eta_2 \end{bmatrix}$$

to represent the parameters $A = (a, b; c, d)$, $\boldsymbol{\tau} = (\tau_1, \tau_2)$ and $\boldsymbol{\eta} = (\eta_1, \eta_2)$ of the 2D OLCT, $\boldsymbol{\tau}$ and $\boldsymbol{\eta}$ are the parameter vectors, where $|\boldsymbol{\tau}|^2 = \tau_1^2 + \tau_2^2$, $|\boldsymbol{\eta}|^2 = \eta_1^2 + \eta_2^2$, and $a, b, c, d, \tau_1, \tau_2, \eta_1, \eta_2$ are real numbers satisfying $ad - bc = 1$. The next derivation provides the definition of the 2D OLCT.

Let $\mathbf{t} = (t_1, t_2)$, $\mathbf{u} = (u_1, u_2)$, $\mathbf{t} \cdot \mathbf{u} = t_1 u_1 + t_2 u_2$, $|\mathbf{t}|^2 = t_1^2 + t_2^2$, $|\mathbf{u}|^2 = u_1^2 + u_2^2$, and $d\mathbf{t} = dt_1 dt_2$, where d stands for differential notation. The 2D OLCT with parameters of $A = (a, b; c, d)$, $\boldsymbol{\tau} = (\tau_1, \tau_2)$, and $\boldsymbol{\eta} = (\eta_1, \eta_2)$ and a signal $f(\mathbf{t})$ is defined by

$$F^{A, \boldsymbol{\tau}, \boldsymbol{\eta}}(\mathbf{u}) = O_L^{A, \boldsymbol{\tau}, \boldsymbol{\eta}}[f(\mathbf{t})](\mathbf{u}) = \begin{cases} \int_{\mathbb{R}^2} f(\mathbf{t}) h_{A, \boldsymbol{\tau}, \boldsymbol{\eta}}(\mathbf{t}, \mathbf{u}) d\mathbf{t}, & b \neq 0 \\ \sqrt{d} e^{i \frac{cd}{2}(\mathbf{u}-\boldsymbol{\tau})^2 + i\boldsymbol{\eta} \cdot \mathbf{u}} f[d(\mathbf{u}-\boldsymbol{\tau})], & b = 0 \end{cases} \quad (3)$$

where the kernel $h_{A, \boldsymbol{\tau}, \boldsymbol{\eta}}(\mathbf{t}, \mathbf{u})$ is

$$h_{A, \boldsymbol{\tau}, \boldsymbol{\eta}}(\mathbf{t}, \mathbf{u}) = K e^{i \left[\frac{a}{2b} |\mathbf{t}|^2 + \frac{1}{b} \mathbf{t} \cdot (\boldsymbol{\tau} - \mathbf{u}) - \frac{1}{b} \mathbf{u} \cdot (d\boldsymbol{\tau} - b\boldsymbol{\eta}) + \frac{d}{2b} |\mathbf{u}|^2 \right]}, \quad (4)$$

$$K = \frac{1}{2\pi b} e^{i \frac{d|\boldsymbol{\tau}|^2}{b}}. \quad (5)$$

Here, $O_L^{A, \boldsymbol{\tau}, \boldsymbol{\eta}}$ denotes the OLCT operator, and a, b, c, d are real numbers satisfying $ad - bc = 1$. When $A = (a, b; c, d)$, $\boldsymbol{\tau} = \mathbf{0}$ and $\boldsymbol{\eta} = \mathbf{0}$, the 2D OLCT reduces to the 2D LCT; when $A = (0, 1; -1, 0)$, $\boldsymbol{\tau} = \mathbf{0}$ and $\boldsymbol{\eta} = \mathbf{0}$, the 2D OLCT can be reduced to the 2D FT.

From the additivity property of the 2D OLCT [8], we can conclude that the inverse operation of the OLCT with parameters $A^{-1} = (d, -b; -c, a)$, $\boldsymbol{\xi} = b\boldsymbol{\eta} - d\boldsymbol{\tau}$, and $\boldsymbol{\gamma} = c\boldsymbol{\tau} - a\boldsymbol{\eta}$ is

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{pmatrix} \mathbf{u} \\ \mathbf{t} \end{pmatrix} + \begin{pmatrix} -d\boldsymbol{\tau} + b\boldsymbol{\eta} \\ c\boldsymbol{\tau} - a\boldsymbol{\eta} \end{pmatrix}$$

The inversion formula of the 2D OLCT is

$$\begin{aligned} f(\mathbf{t}) &= O_L^{A^{-1}, \boldsymbol{\xi}, \boldsymbol{\gamma}}[F^{A, \boldsymbol{\tau}, \boldsymbol{\eta}}(\mathbf{u})](\mathbf{t}) \\ &= C \frac{\sigma}{2\pi b} \int_{\mathbb{R}^2} F^{A, \boldsymbol{\tau}, \boldsymbol{\eta}}(\mathbf{u}) e^{i \left[-\frac{d}{2b} |\mathbf{u}|^2 + \frac{1}{b} \mathbf{u} \cdot \mathbf{t} - \frac{1}{b} \mathbf{u} \cdot (b\boldsymbol{\eta} - d\boldsymbol{\tau}) + \frac{(bc-ad)}{b} \mathbf{t} \cdot \boldsymbol{\tau} - \frac{a}{2b} |\mathbf{t}|^2 \right]} d\mathbf{u}, \end{aligned} \quad (6)$$

where d stands for differential notation, $C = e^{\frac{i}{2} [cd\boldsymbol{\tau}^2 - 2ad\boldsymbol{\tau} \cdot \boldsymbol{\eta} + ab\boldsymbol{\eta}^2]}$ and $\sigma = -e^{-i \frac{a[b\boldsymbol{\eta} - d\boldsymbol{\tau}]^2}{b}}$.

The definition for the case when $b = 0$ is the limit of the integral in (3); for the case when $b \neq 0$ is $|b| \rightarrow 0$, the OLCT is simply a time-scaled version multiplied by a linear chirp. Therefore, we shall confine our attention to the 2D OLCT for $b \neq 0$. Without loss of generality, we assume $b > 0$ in the following sections.

3. 2D offset linear canonical transform and offset linear canonical Hankel transform in polar coordinates

3.1. 2D offset linear canonical transform and offset linear canonical Hankel transform

We assume that the function— $f(r, \theta)$ satisfies the following conditions:

1. The function— $f(r, \theta)$ satisfies the Dirichlet conditions, i.e.,
 - (i) The function $f(r)$ is piecewise continuous in $r \in (0, \infty)$, and its variation is bounded in every subinterval $[r_1, r_2]$.
 - (ii) The integral is

$$\int_0^{\infty} \sqrt{r} |f(r)| dr < \infty,$$

where d stands for differential notation.

- (iii) The Fourier-Bessel series of $f(r)$ converges to $f(r)$ at every continuity point of $f(r)$ and to $\frac{1}{2} [f(r+0) + f(r-0)]$ at every discontinuity point of $f(r)$.
2. The function $f(r, \theta)$ is angularly periodic in 2π , i.e.,

$$f(r, \theta) = f(r, \theta + 2\pi k) \quad k = \pm 1, \pm 2, \dots$$

and lends itself to a Fourier series expansion of the form

$$f(r, \theta) = \sum_{n=-\infty}^{+\infty} f_n(r) e^{in\theta}. \quad (7)$$

Here, we make the following symbolic regulations:

The function $f(r, \theta)$ is defined; its 2D FT $F(\rho, \phi)$, the 2D OFT $F^{\tau, \eta}(\rho, \phi)$ and the 2D OLCT $F^{A, \tau, \eta}(\rho, \phi)$ have the following notations:

$$\begin{aligned} f(r \cos \theta, r \sin \theta) &\triangleq f(r, \theta), \\ F(\rho \cos \phi, \rho \sin \phi) &\triangleq F(\rho, \phi), \\ F^{\tau, \eta}(\rho \cos \phi, \rho \sin \phi) &\triangleq F^{\tau, \eta}(\rho, \phi), \\ F^{A, \tau, \eta}(\rho \cos \phi, \rho \sin \phi) &\triangleq F^{A, \tau, \eta}(\rho, \phi). \end{aligned}$$

By changing from Cartesian coordinates to polar coordinates related to the general functions of $f(\mathbf{t})$, the 2D FT $F(\mathbf{u})$, 2D OFT $F^{\tau, \eta}(\mathbf{u})$ and 2D OLCT $F^{A, \tau, \eta}(\mathbf{u})$ are obtained.

In polar coordinates, we use (r, θ) in physical polar coordinates to represent $\mathbf{t} = (t_1, t_2)$, and the polar coordinates frequency vector (ρ, ϕ) in the OLCT domain to represent $\mathbf{u} = (u_1, u_2)$. According to the 2D OLCT, using the polar coordinates $t_1 = r \cos \theta$, $t_2 = r \sin \theta$, $u_1 = \rho \cos \phi$, and $u_2 = \rho \sin \phi$, we can obtain the definition of the 2D OLCT in polar coordinates as follows:

Definition 1 (2D OLCT). Assuming that the function $f(r, \theta)$ is angularly periodic in 2π , its 2D OLCT with parameters of $A = (a, b; c, d)$, $\tau = (\tau_1, \tau_2)$, and $\eta = (\eta_1, \eta_2)$ in polar coordinates is defined by

$$F^{A, \tau, \eta}(\rho, \phi) = O_L^{A, \tau, \eta}[f](\rho, \phi) = \int_0^{+\infty} \int_0^{2\pi} f(r, \theta) P_{A, \tau, \eta}(r, \theta; \rho, \phi) r dr d\theta, \quad (8)$$

where d stands for differential notation, $P_{A, \tau, \eta}(r, \theta; \rho, \phi)$ denotes the 2D OLCT kernel in polar coordinates and is given by

$$P_{A, \tau, \eta}(r, \theta; \rho, \phi) = K e^{i \left[\frac{a}{2b} r^2 - \frac{r\rho}{b} \cos(\theta - \phi) + \frac{r|\tau|}{b} \sin(\theta + \varphi_1) - \frac{\rho|d\tau - b\eta|}{b} \sin(\phi + \varphi_2) + \frac{d}{2b} \rho^2 \right]}. \quad (9)$$

Here, τ and η are the parameter vectors, $|\tau|^2 = \tau_1^2 + \tau_2^2$, $|\eta|^2 = \eta_1^2 + \eta_2^2$, $a, b, c, d, \tau_1, \tau_2, \eta_1, \eta_2$ are real numbers satisfying $ad - bc = 1$, $0 \leq \theta, \phi \leq 2\pi$, $\tan \varphi_1 = \frac{\tau_1}{\tau_2}$, $\tan \varphi_2 = \frac{d\tau_1 - b\eta_1}{d\tau_2 - b\eta_2}$, $\tau_2 \neq 0$, $d\tau_2 - b\eta_2 \neq 0$, and K is given by (5).

In terms of polar coordinates, the OLCT operation transforms the spatial position radius and angle (r, θ) to the frequency radius and angle (ρ, ϕ) . The inversion formula of the 2D OLCT in polar coordinates takes the following form:

$$f(r, \theta) = O_L^{A^{-1}, \xi, \lambda}[F^{A, \tau, \eta}](r, \theta), \quad (10)$$

where $A^{-1} = (d, -b; -c, a)$, $\xi = b\eta - d\tau$, and $\lambda = c\tau - a\eta$.

Remark 1. When $A = (0, 1; -1, 0)$, $\tau = \mathbf{0}$, and $\eta = \mathbf{0}$, the 2D OLCT can be reduced to the 2D FT

$$\begin{aligned} F(\rho, \phi) &= \mathcal{F}[f](\rho, \phi) \\ &= \frac{1}{2\pi} \int_0^{+\infty} \int_0^{2\pi} f(r, \theta) e^{-ir\rho \cos(\theta-\phi)} r dr d\theta. \end{aligned} \quad (11)$$

It follows that a relation exists between the 2D OLCT and the 2D FT

$$F^{A, \tau, \eta}(\rho, \phi) = \frac{\ell}{b} e^{i \left[\frac{d}{2b} \rho^2 - \frac{\rho |d\tau - b\eta|}{b} \sin(\phi + \varphi_2) \right]} F[\tilde{f}] \left(\frac{\rho}{b}, \phi \right), \quad (12)$$

where $\tilde{f}(r, \theta) = e^{i \left[\frac{a}{2b} r^2 + \frac{r|\tau|}{b} \sin(\theta + \varphi_1) \right]} f(r, \theta)$, φ_1 and φ_2 are given by (9), and

$$\ell = e^{i \frac{d|\tau|^2}{b}}. \quad (13)$$

Remark 2. When $A = (0, 1; -1, 0)$, $\tau = (\tau_1, \tau_2)$, and $\eta = (\eta_1, \eta_2)$, where $\tau_1, \tau_2, \eta_1, \eta_2$ are real numbers. Then the 2D OLCT can be reduced to the 2D offset Fourier transform (OFT) [8,9]

$$\begin{aligned} F^{\tau, \eta}(\rho, \phi) &= \mathcal{F}^{\tau, \eta}[f](\rho, \phi) \\ &= \frac{1}{2\pi} \int_0^{+\infty} \int_0^{2\pi} f(r, \theta) e^{i[r|\tau| \sin(\theta + \varphi_1) - \rho|\eta| \sin(\phi + \varphi_2) - r\rho \cos(\theta - \phi)]} r dr d\theta, \end{aligned} \quad (14)$$

where d stands for differential notation, φ_1 and φ_2 are given by (9).

It follows that a relation exists between the 2D OLCT and the 2D OFT

$$F^{A, \tau, \eta}(\rho, \phi) = \frac{\ell}{b} e^{i \left[\frac{d}{2b} \rho^2 - \frac{\rho(|d\tau - b\eta| - b|\eta|)}{b} \sin(\phi + \varphi_2) \right]} F^{\tau, \eta}[\tilde{f}](\rho, \phi), \quad (15)$$

where $\tilde{f}(r, \theta) = e^{i \frac{a}{2b} r^2} f\left(\frac{r}{b}, \theta\right)$, φ_1 and φ_2 are given by (9), and ℓ is given by (13).

Remark 3. When $A = (a, b; c, d)$, $\tau = \mathbf{0}$, and $\eta = \mathbf{0}$, where a, b, c, d are real numbers satisfying $ad - bc = 1$. Then the 2D OLCT can be reduced to the 2D LCT [25]. Definition 1 is a more general extension of reference [25], which multiple parameters reflect the flexibility of the proposed transformation.

Next, we discuss the definition of the OLCHT in polar coordinates. Inspired by Reference [25–27] and under the premise that the transformation function has rotational symmetry, we obtain the OLCHT from the OLCT.

Lemma 1 (OLCHT). The n th-order OLCHT of the parameter matrices of $A = (a, b; c, d)$, $\tau = (\tau_1, \tau_2)$, and $\eta = (\eta_1, \eta_2)$ is defined by

$$H_n^{A, \tau, \eta}[f](\rho) = \frac{i^n w_1 \ell e^{im(\varphi_1 - \varphi_2)}}{b} e^{i \frac{d}{2b} \rho^2} \int_0^{+\infty} w_2 e^{i \frac{a}{2b} r^2} f(r) J_n \left(\frac{r\rho}{b} \right) r dr, \quad (16)$$

where J_n is the n th-order Bessel function of the first kind with order $n \geq -\frac{1}{2}$, d stands for differential notation, φ_1 and φ_2 are given by (9), ℓ is given by (13), and

$$w_1 = \sum_{m=-\infty}^{+\infty} J_m \left(\frac{\rho |d\tau - b\eta|}{b} \right), \quad w_2 = \sum_{m=-\infty}^{+\infty} J_m \left(\frac{r|\tau|}{b} \right). \quad (17)$$

Here, τ and η are the parameter vectors, $|\tau|^2 = \tau_1^2 + \tau_2^2$, $|\eta|^2 = \eta_1^2 + \eta_2^2$, $a, b, c, d, \tau_1, \tau_2, \eta_1, \eta_2$ are real numbers satisfying $ad - bc = 1$.

Proof. From (8), it follows that

$$\begin{aligned} F^{A, \tau, \eta}(\rho, \phi) &= \frac{\ell}{2\pi b} \int_0^{+\infty} \int_0^{2\pi} f(r, \theta) e^{i \left[\frac{a}{2b} r^2 - \frac{r\rho}{b} \cos(\theta - \phi) + \frac{d}{2b} \rho^2 \right]} \\ &\quad \times e^{i \left[\frac{r|\tau|}{b} \sin(\theta + \varphi_1) - \frac{\rho |d\tau - b\eta|}{b} \sin(\phi + \varphi_2) \right]} r dr d\theta, \end{aligned} \quad (18)$$

where φ_1 and φ_2 are given by (9), and ℓ is given by (13).

In view of Formula [28]

$$e^{-it \sin \theta} = \sum_{m=-\infty}^{+\infty} J_m(t) e^{-im\theta}. \quad (19)$$

Therefore, we can obtain

$$e^{i\frac{r|\tau|}{b}\sin(\theta+\varphi_1)} = \sum_{m=-\infty}^{+\infty} J_m\left(\frac{r|\tau|}{b}\right) e^{im(\theta+\varphi_1)}, \quad (20)$$

$$e^{-i\frac{\rho|d\tau-b\eta|}{b}\sin(\phi+\varphi_2)} = \sum_{m=-\infty}^{+\infty} J_m\left(\frac{\rho|d\tau-b\eta|}{b}\right) e^{-im(\phi+\varphi_2)}. \quad (21)$$

When $f(r, \theta)$ is rotational symmetry, there is

$$f(r, \theta) = f(r) e^{ik\theta}. \quad (22)$$

Let

$$B = \frac{1}{2\pi} \int_0^{2\pi} f(r, \theta) e^{i\left[\frac{r|\tau|}{b}\sin(\theta+\varphi_1) - \frac{\rho|d\tau-b\eta|}{b}\sin(\phi+\varphi_2) - \frac{r\rho}{b}\cos(\theta-\phi)\right]} d\theta. \quad (23)$$

Using (20), (21) and (22), we can derive the following result

$$\begin{aligned} B &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{m=-\infty}^{+\infty} \sum_{m=-\infty}^{+\infty} J_m\left(\frac{r|\tau|}{b}\right) J_m\left(\frac{\rho|d\tau-b\eta|}{b}\right) \\ &\quad \times e^{im\theta - im\phi + im(\varphi_1 - \varphi_2) - i\frac{r\rho}{b}\cos(\theta-\phi)} e^{ik\theta} f(r) d\theta. \end{aligned} \quad (24)$$

According to Formula [24,28]

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n\theta - x\sin\theta)} d\theta. \quad (25)$$

We have

$$\begin{aligned} &\frac{1}{2\pi} \int_0^{2\pi} e^{im\theta - im\phi + im(\varphi_1 - \varphi_2) - i\frac{r\rho}{b}\cos(\theta-\phi)} \cdot e^{ik\theta} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} e^{im\theta - im\phi + im(\varphi_1 - \varphi_2) - i\frac{r\rho}{b}\sin(\frac{\pi}{2} + \theta - \phi)} \cdot e^{ik\theta} d\theta \\ &= e^{im(\varphi_1 - \varphi_2) - in(\frac{\pi}{2} + \theta - \phi) + im\theta - im\phi + ik\theta} \frac{1}{2\pi} \int_0^{2\pi} e^{i\left[n(\frac{\pi}{2} + \theta - \phi) - \frac{r\rho}{b}\sin(\frac{\pi}{2} + \theta - \phi)\right]} d\theta \\ &= J_n\left(\frac{r\rho}{b}\right) e^{i\left[m(\varphi_1 - \varphi_2) + (m+k-n)\theta + (n-m)\phi - \frac{\pi}{2}n\right]}. \end{aligned} \quad (26)$$

By substituting $k = n - m$ in the above expression, we obtain

$$\begin{aligned} &\frac{1}{2\pi} \int_0^{2\pi} e^{im\theta - im\phi + im(\varphi_1 - \varphi_2) - i\frac{r\rho}{b}\cos(\theta-\phi)} \cdot e^{ik\theta} d\theta \\ &= J_n\left(\frac{r\rho}{b}\right) e^{ik\phi - i\frac{\pi}{2}n + im(\varphi_1 - \varphi_2)}. \end{aligned} \quad (27)$$

Hence

$$\begin{aligned} F^{A,\tau,\eta}(\rho, \phi) &= \frac{i^n \ell e^{im(\varphi_1 - \varphi_2)}}{b} \sum_{m=-\infty}^{+\infty} J_m\left(\frac{\rho|d\tau-b\eta|}{b}\right) e^{i\frac{d}{2b}\rho^2} \\ &\quad \times \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} J_m\left(\frac{r|\tau|}{b}\right) e^{i\frac{a}{2b}r^2} J_n\left(\frac{r\rho}{b}\right) f(r) r dr e^{ik\phi}. \end{aligned} \quad (28)$$

It is known from (28) that a function with rotational symmetry under the OLCCT maintains its symmetry. Thus, we can write

$$F^{A,\tau,\eta}(\rho, \phi) = F^{A,\tau,\eta}(\rho) e^{ik\phi}. \quad (29)$$

Therefore

$$\begin{aligned}
H_n^{A, \tau, \eta}[f](\rho) &= F^{A, \tau, \eta}(\rho) \\
&= \frac{i^n \ell e^{im(\varphi_1 - \varphi_2)}}{b} \sum_{m=-\infty}^{+\infty} J_m\left(\frac{\rho |d\tau - b\eta|}{b}\right) e^{i \frac{d}{2b} \rho^2} \\
&\quad \times \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} J_m\left(\frac{r|\tau|}{b}\right) e^{i \frac{a}{2b} r^2} f(r) J_n\left(\frac{r\rho}{b}\right) r dr.
\end{aligned} \tag{30}$$

Let $w_1 = \sum_{m=-\infty}^{+\infty} J_m\left(\frac{\rho |d\tau - b\eta|}{b}\right)$, $w_2 = \sum_{m=-\infty}^{+\infty} J_m\left(\frac{r|\tau|}{b}\right)$; hence

$$H_n^{A, \tau, \eta}[f](\rho) = \frac{i^n \ell e^{im(\varphi_1 - \varphi_2)}}{b} w_1 e^{i \frac{d}{2b} \rho^2} \int_0^{+\infty} w_2 e^{i \frac{a}{2b} r^2} f(r) J_n\left(\frac{r\rho}{b}\right) r dr. \tag{31}$$

The proof is completed. $\square$

The inversion formula of the n th-order OLCHT is

$$f(r) = H_n^{-A^{-1}, -\xi, -\gamma} \left[H_n^{A, \tau, \eta}[f] \right](r), \tag{32}$$

where $A^{-1} = (d, -b; -c, a)$, $\xi = b\eta - d\tau$, and $\lambda = c\tau - a\eta$.

Remark 4. It is evident that the n th-order OLCHT and its inverse with $A = (0, 1; -1, 0)$, $\tau = \mathbf{0}$, and $\eta = \mathbf{0}$ reduces to the conventional n th-order Hankel transform (HT) [29]

$$H_n[f](\rho) = \int_0^{+\infty} f(r) J_n(\rho r) r dr, \tag{33}$$

and the corresponding inversion formula

$$f(r) = H_n[H_n[f]](r), \tag{34}$$

respectively.

Remark 5. It is evident that the n th-order OLCHT with $A = (a, b; c, d)$, $\tau = \mathbf{0}$, and $\eta = \mathbf{0}$ reduces to the conventional n th-order linear canonical Hankel transform (LCHT) [25]. It also shows that the OLCHT is a further extension of the LCHT. The diversity of its parameters determines that the OLCHT has greater freedom and flexibility, and its mathematical derivation is more complex.

3.2. Relationship between the 2D OLCT and the OLCHT

Under the condition of rotational symmetry, the OLCHT can be deduced from the OLCT, and there is a close relationship between them. The purpose of this subsection is to extend the above relationship to prepare for the OLCT domain.

Let the function $f(r, \theta)$ and its 2D FT $F(\rho, \phi)$ satisfy the Dirichlet conditions. Since they can be angularly periodic with a period of 2π , their Fourier series are well-defined as

$$f(r, \theta) = \sum_{n=-\infty}^{+\infty} f_n(r) e^{in\theta}, \tag{35}$$

$$F(\rho, \phi) = \sum_{n=-\infty}^{+\infty} F_n(\rho) e^{in\phi}. \tag{36}$$

Notably, the n th term in the Fourier series of the primitive functions and their 2D FT versions $f_n(r)$ and $F_n(\rho)$ form a n th-order HT pair [29–31]

$$F_n(\rho) = i^{-n} H_n[f_n](\rho), \tag{37}$$

$$f_n(r) = i^n H_n[F_n](r). \tag{38}$$

From the above equations (16) and (33), we can obtain the relationship between the OLCHT and the HT; it follows that

$$H_n^{A, \tau, \eta}[f](\rho) = \frac{i^n w_1 \ell e^{im(\varphi_1 - \varphi_2)}}{b} e^{i \frac{d}{2b} \rho^2} H_n[\tilde{f}]\left(\frac{\rho}{b}\right), \tag{39}$$

where φ_1 and φ_2 are given by (9), ℓ is given by (13), and w_1 and w_2 are given by (17),

$$\tilde{f}(r) = w_2 e^{i \frac{a}{2b} r^2} f(r). \tag{40}$$

Remark 6. If it is assumed that f is radially symmetric, then it can be written as a function of only r and can thus be taken out of the integration over the angular coordinate.

Remark 7. When the function $f(r, \theta)$ is not radially symmetric and is a function of both r and θ , the preceding result can be generalized. Since $f(r, \theta)$ depends on the angle θ , it can be expanded into a Fourier series with (35) and (36).

Next, we show the relation between the 2D OLCT and the OLCHT.

Theorem 1. The function $f(r, \theta)$ satisfies the Dirichlet conditions for every fixed $r \in [0, \infty]$, is angularly periodic in 2π , and has the following Fourier expansion

$$f(r, \theta) = \sum_{n=-\infty}^{+\infty} f_n(r) e^{in\theta}. \quad (41)$$

Then, the Fourier series expansion of the 2D OLCT of $f(r, \theta)$ has the form

$$F^{A, \tau, \eta}(\rho, \phi) = \sum_{n=-\infty}^{+\infty} H_n^{A, \tau, \eta}[f_n](\rho) e^{in\phi}. \quad (42)$$

Proof. The 2D OLCT of $f(r, \theta)$, as expanded in (41), takes the form

$$\begin{aligned} F^{A, \tau, \eta}(\rho, \phi) &= \sum_{n=-\infty}^{+\infty} \int_0^{+\infty} \int_0^{2\pi} e^{in\theta} f_n(r) P_{A, \tau, \eta}(r, \theta; \rho, \phi) r dr d\theta \\ &= \sum_{n=-\infty}^{+\infty} \int_0^{+\infty} \underbrace{e^{i \left[\frac{r|\tau|}{b} \sin(\theta + \varphi_1) - \frac{\rho|d\tau - b\eta|}{b} \sin(\phi + \varphi_2) - \frac{r\rho}{b} \cos(\theta - \phi) \right]}_{D(r, \theta; \rho, \phi)} e^{in\theta} \\ &\quad \times \frac{\ell}{2\pi b} e^{i \left(\frac{a}{2b} r^2 + \frac{d}{2b} \rho^2 \right)} \int_0^{2\pi} f_n(r) r dr d\theta, \end{aligned} \quad (43)$$

where d stands for differential notation, φ_1 and φ_2 are given by (9), and ℓ is given by (13).

According to the popular exponential expansion formula [28], we have

$$e^{i \frac{r\rho}{b} \cos(\theta - \phi)} = \sum_{v=-\infty}^{+\infty} i^v J_v \left(\frac{r\rho}{b} \right) e^{-iv(\theta - \phi)}. \quad (44)$$

Using (20), (21) and (44), we get

$$\begin{aligned} D(r, \theta; \rho, \phi) &= \sum_{v=-\infty}^{+\infty} i^v J_v \left(\frac{r\rho}{b} \right) w_1 w_2 e^{im\theta - im\phi + im(\varphi_1 - \varphi_2) - iv(\theta - \phi) + in\theta} \\ &= \sum_{v=-\infty}^{+\infty} i^v J_v \left(\frac{r\rho}{b} \right) w_1 w_2 e^{i(m-v+n)\theta - i(m-v)\phi + im(\varphi_1 - \varphi_2)}, \end{aligned} \quad (45)$$

where w_1 and w_2 are given by (17).

For $n = v - m$, (45) can be written as

$$D(r, \theta; \rho, \phi) = \sum_{v=-\infty}^{+\infty} i^v J_v \left(\frac{r\rho}{b} \right) w_1 w_2 e^{in\phi + im(\varphi_1 - \varphi_2)}. \quad (46)$$

Hence

$$\begin{aligned} F^{A, \tau, \eta}(\rho, \phi) &= \sum_{n=-\infty}^{+\infty} \int_0^{+\infty} \frac{\ell}{2\pi b} e^{i \left(\frac{a}{2b} r^2 + \frac{d}{2b} \rho^2 \right)} \sum_{v=-\infty}^{+\infty} i^v J_v \left(\frac{r\rho}{b} \right) w_1 w_2 \\ &\quad \times e^{in\phi + im(\varphi_1 - \varphi_2)} f_n(r) r dr \int_0^{2\pi} d\theta \\ &= \sum_{n=-\infty}^{+\infty} \frac{i^n \ell e^{im(\varphi_1 - \varphi_2)}}{b} w_1 e^{i \frac{d}{2b} \rho^2} \int_0^{+\infty} w_2 e^{i \frac{a}{2b} r^2} J_n \left(\frac{r\rho}{b} \right) f_n(r) e^{in\phi} r dr \\ &= \sum_{n=-\infty}^{+\infty} H_n^{A, \tau, \eta}[f_n](\rho) e^{in\phi}, \end{aligned} \quad (47)$$

where

$$w_1 = \sum_{m=-\infty}^{+\infty} J_m \left(\frac{\rho |d\tau - b\eta|}{b} \right), w_2 = \sum_{m=-\infty}^{+\infty} J_m \left(\frac{r|\tau|}{b} \right).$$

The proof is completed. $\square$

Through the above discussions, the Fourier series expansion of the 2D OLCT $F^{A,\tau,\eta}(\rho, \phi)$ has the form

$$F^{A,\tau,\eta}(\rho, \phi) = \sum_{n=-\infty}^{+\infty} F_n^{A,\tau,\eta}(\rho) e^{in\phi}. \quad (48)$$

According to Theorem 1 and (48), we obtain

$$F_n^{A,\tau,\eta}(\rho) = H_n^{A,\tau,\eta}[f_n](\rho). \quad (49)$$

4. Properties of the 2D OLCT in polar coordinates

The 1D OLCT has many important properties, such as space shift and phase shift properties. Similarly, the 2D OLCT also has some important qualities. In this section, we give two theorems of the 2D OLCT in polar coordinates, namely, the spatial shift theorem and the convolution theorem.

4.1. Spatial shift theorem

The spatial shift property of the 1D OLCT is provided in [6]. Below, we give the spatial shift theorem of the 2D OLCT in polar coordinates.

Theorem 2 (Spatial shift theorem). Let $\mathbf{t}_0 = (t_3, t_4)$, using polar coordinates, $t_3 = r_0 \cos \theta_0$, $t_4 = r_0 \sin \theta_0$. Then, we have

$$\begin{aligned} f(\mathbf{t} - \mathbf{t}_0) &= \frac{\sigma C}{b} \sum_{n=-\infty}^{+\infty} \gamma(r, \theta; r_0, \theta_0) e^{in(\theta - \theta_0 - \varphi_2)} H_n^{A,\tau,\eta}[f_n](\rho) e^{-i\frac{d}{2b}\rho^2} \\ &\times \int_0^{+\infty} J_n\left(\frac{r\rho}{b}\right) J_n\left(\frac{r_0\rho}{b}\right) J_n\left(\frac{\rho|b\eta - d\tau|}{b}\right) \rho d\rho, \end{aligned} \quad (50)$$

where (r_0, θ_0) is the spatial translation position radius and angle, d stands for differential notation, σ and C are given by (6), $\tan \varphi_2 = \frac{d\tau_1 - b\eta_1}{d\tau_2 - b\eta_2}$, $d\tau_2 - b\eta_2 \neq 0$, and

$$\gamma(r, \theta; r_0, \theta_0) = e^{i\left[-\frac{a}{2b}(r^2 + r_0^2) + \frac{arr_0}{b} \cos(\theta - \theta_0) + \frac{(bc-ad)r|\tau|}{b} \sin(\theta + \varphi_1) - \frac{(bc-ad)r_0|\tau|}{b} \sin(\theta_0 + \varphi_1)\right]}.$$

Here, τ and η are the parameter vectors, $|\tau|^2 = \tau_1^2 + \tau_2^2$, $|\eta|^2 = \eta_1^2 + \eta_2^2$, $a, b, c, d, \tau_1, \tau_2, \eta_1, \eta_2$ are real numbers satisfying $ad - bc = 1$.

Proof. According to (6), it follows that

$$f(\mathbf{t}) = C \frac{\sigma}{2\pi b} \int_{\mathbb{R}^2} F^{A,\tau,\eta}(\mathbf{u}) e^{i\left[-\frac{d}{2b}|\mathbf{u}|^2 + \frac{1}{b}\mathbf{u} \cdot \mathbf{t} - \frac{1}{b}\mathbf{u} \cdot (b\eta - d\tau) + \frac{(bc-ad)}{b}\mathbf{t} \cdot \tau - \frac{a}{2b}|\mathbf{t}|^2\right]} d\mathbf{u},$$

where $\sigma = -e^{-i\frac{a|b\eta - d\tau|^2}{b}}$ and $C = e^{\frac{i}{2}[cd\tau^2 - 2ad\tau \cdot \eta + ab\eta^2]}$.

Hence, we obtain

$$\begin{aligned} f(\mathbf{t} - \mathbf{t}_0) &= C \frac{\sigma}{2\pi b} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F^{A,\tau,\eta}(\mathbf{u}) e^{i\left[-\frac{d}{2b}|\mathbf{u}|^2 + \frac{1}{b}\mathbf{u} \cdot (\mathbf{t} - \mathbf{t}_0)\right]} \\ &\times e^{i\left[-\frac{1}{b}\mathbf{u} \cdot (b\eta - d\tau) + \frac{(bc-ad)}{b}(\mathbf{t} - \mathbf{t}_0) \cdot \tau - \frac{a}{2b}|\mathbf{t} - \mathbf{t}_0|^2\right]} d\mathbf{u}. \end{aligned} \quad (51)$$

Using polar coordinates $t_1 = r \cos \theta$, $t_2 = r \sin \theta$, $t_3 = r_0 \cos \theta_0$, $t_4 = r_0 \sin \theta_0$, $u_1 = \rho \cos \phi$, and $u_2 = \rho \sin \phi$, we obtain

$$\begin{aligned} E &\triangleq e^{i\left[-\frac{d}{2b}|\mathbf{u}|^2 + \frac{1}{b}\mathbf{u} \cdot (\mathbf{t} - \mathbf{t}_0) - \frac{1}{b}\mathbf{u} \cdot (b\eta - d\tau) + \frac{(bc-ad)}{b}(\mathbf{t} - \mathbf{t}_0) \cdot \tau - \frac{a}{2b}|\mathbf{t} - \mathbf{t}_0|^2\right]} \\ &= e^{i\left[-\frac{d}{2b}\rho^2 - \frac{a}{2b}(r^2 + r_0^2) + \frac{arr_0}{b} \cos(\theta - \theta_0) - \frac{\rho|b\eta - d\tau|}{b} \sin(\phi + \varphi_2) + \frac{(bc-ad)r|\tau|}{b} \sin(\theta + \varphi_1)\right]} \\ &\times e^{i\left[-\frac{(bc-ad)r_0|\tau|}{b} \sin(\theta_0 + \varphi_1) + \frac{\rho r}{b} \cos(\theta - \phi) - \frac{\rho r_0}{b} \cos(\theta_0 - \phi)\right]}, \end{aligned} \quad (52)$$

where $\tan \varphi_1 = \frac{\tau_1}{\tau_2}$, $\tan \varphi_2 = \frac{d\tau_1 - b\eta_1}{d\tau_2 - b\eta_2}$, $\tau_2 \neq 0$, and $d\tau_2 - b\eta_2 \neq 0$.

According to the popular exponential expansion formula in (19), by splitting each item of E , we have

$$e^{i\frac{r\rho}{b}\cos(\theta-\phi)} = \sum_{m=-\infty}^{+\infty} i^m J_m\left(\frac{r\rho}{b}\right) e^{im(\theta-\phi)}, \quad (53)$$

$$e^{-i\frac{r_0\rho}{b}\cos(\theta_0-\phi)} = \sum_{m=-\infty}^{+\infty} i^{-m} J_m\left(\frac{r_0\rho}{b}\right) e^{-im(\theta_0-\phi)}, \quad (54)$$

and

$$e^{-i\frac{\rho|b\eta-d\tau|}{b}\sin(\phi+\varphi_2)} = \sum_{n=-\infty}^{+\infty} J_n\left(\frac{\rho|b\eta-d\tau|}{b}\right) e^{-in(\phi+\varphi_2)}. \quad (55)$$

Using (42), (52), (53), (54) and (55), we obtain

$$\begin{aligned} & f(\mathbf{t} - \mathbf{t}_0) \\ &= C \frac{\sigma}{2\pi b} \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} J_m\left(\frac{r\rho}{b}\right) J_m\left(\frac{r_0\rho}{b}\right) \sum_{n=-\infty}^{+\infty} J_n\left(\frac{\rho|b\eta-d\tau|}{b}\right) \\ &\quad \times \sum_{n=-\infty}^{+\infty} H_n^{A,\tau,\eta}[f_n](\rho) e^{im(\theta-\theta_0)-in\varphi_2} \rho d\rho \int_0^{2\pi} d\phi \\ &\quad \times e^{i\left[-\frac{d}{2b}\rho^2 - \frac{a}{2b}(r^2+r_0^2) + \frac{arr_0}{b}\cos(\theta-\theta_0) + \frac{(bc-ad)r|\tau|}{b}\sin(\theta+\varphi_1) - \frac{(bc-ad)r_0|\tau|}{b}\sin(\theta_0+\varphi_1)\right]}. \end{aligned} \quad (56)$$

Letting $n = m$, we obtain

$$\begin{aligned} & f(\mathbf{t} - \mathbf{t}_0) \\ &= C \frac{\sigma}{b} \int_0^{+\infty} \sum_{n=-\infty}^{+\infty} J_n\left(\frac{r\rho}{b}\right) J_n\left(\frac{r_0\rho}{b}\right) J_n\left(\frac{\rho|b\eta-d\tau|}{b}\right) \\ &\quad \times e^{i\left[-\frac{d}{2b}\rho^2 - \frac{a}{2b}(r^2+r_0^2) + \frac{arr_0}{b}\cos(\theta-\theta_0) + \frac{(bc-ad)r|\tau|}{b}\sin(\theta+\varphi_1) - \frac{(bc-ad)r_0|\tau|}{b}\sin(\theta_0+\varphi_1)\right]} \\ &\quad \times H_n^{A,\tau,\eta}[f_n](\rho) e^{in(\theta-\theta_0-\varphi_2)} \rho d\rho. \end{aligned} \quad (57)$$

Changing the integral and summation signs, we have

$$\begin{aligned} f(\mathbf{t} - \mathbf{t}_0) &= \frac{\sigma C}{b} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta_0-\varphi_2)} \times e^{i\left[\frac{d}{2b}\rho^2 - \frac{a}{2b}(r^2+r_0^2)\right]} \\ &\quad \times e^{i\left[-\frac{arr_0}{b}\cos(\theta-\theta_0) + \frac{(bc-ad)r|\tau|}{b}\sin(\theta+\varphi_1) - \frac{(bc-ad)r_0|\tau|}{b}\sin(\theta_0+\varphi_1)\right]} \\ &\quad \times \int_0^{+\infty} H_n^{A,\tau,\eta}[f_n](\rho) J_n\left(\frac{r\rho}{b}\right) J_n\left(\frac{r_0\rho}{b}\right) J_n\left(\frac{\rho|b\eta-d\tau|}{b}\right) \rho d\rho, \end{aligned} \quad (58)$$

where $\sigma = -e^{-i\frac{a|b\eta-d\tau|^2}{b}}$.

The proof is completed. $\square$

4.2. Convolution theorem

The convolution concept of the 2D OLCT in polar coordinates is as follows:

Definition 2. The convolution operation $*^{A,\tau,\eta}$ of the 2D OLCT in polar coordinates for two functions $f(r, \theta)$ and $g(r, \theta)$ is defined by

$$\begin{aligned} (f *^{A,\tau,\eta} g)(r, \theta) &= \frac{e^{-i\left[\frac{a}{2b}r^2 + \frac{r|\tau|}{b}\sin(\theta+\varphi_1)\right]}}{2\pi} \int_0^{+\infty} e^{i\left[\frac{a}{2b}r_0^2 + \frac{r_0|\tau|}{b}\sin(\theta_0+\varphi_1)\right]} g(r_0, \theta_0) \\ &\quad \times \int_0^{2\pi} F^{-1}\left[e^{-i\rho r_0\cos(\phi-\theta_0)} F^{A,\tau,\eta}(b\rho, \phi)\right](r, \theta) r_0 dr_0 d\theta_0, \end{aligned} \quad (59)$$

where τ and η are the parameter vectors, $|\tau|^2 = \tau_1^2 + \tau_2^2$, $|\eta|^2 = \eta_1^2 + \eta_2^2$, $a, b, c, d, \tau_1, \tau_2, \eta_1, \eta_2$ are real numbers satisfying $ad - bc = 1$, d stands for differential notation, $\tan \varphi_1 = \frac{\tau_1}{\tau_2}$ and $\tau_2 \neq 0$.

In the following, we obtain the convolution theorem of the 2D OLCT in polar coordinates.

Theorem 3 (Convolution theorem). Let $z(r, \theta) = (f *^{A, \tau, \eta} g)(r, \theta)$. Then, there is a relation

$$Z^{A, \tau, \eta}(\rho, \phi) = F^{A, \tau, \eta}(\rho, \phi) G^{A, \tau, \eta}(\rho, \phi), \quad (60)$$

where $Z^{A, \tau, \eta}(\rho, \phi)$, $F^{A, \tau, \eta}(\rho, \phi)$ and $G^{A, \tau, \eta}(\rho, \phi)$ denote the 2D OLCT in polar coordinates of the functions $z(r, \theta)$, $f(r, \theta)$ and $g(r, \theta)$, respectively.

Proof. In view of (42), we obtain

$$e^{-i\rho r_0 \cos(\phi - \theta_0)} = \sum_{m=-\infty}^{+\infty} i^{-m} J_m(\rho r_0) e^{-im\theta_0} e^{im\phi}. \quad (61)$$

By (11) and (61), it follows that

$$\begin{aligned} & F^{-1} \left[e^{-i\rho r_0 \cos(\phi - \theta_0)} F^{A, \tau, \eta}(b\rho, \phi) \right] (r, \theta) \\ &= \frac{1}{2\pi} \int_0^{+\infty} \int_0^{2\pi} \sum_{m=-\infty}^{+\infty} i^{-m} J_m(\rho r_0) e^{-im\theta_0} e^{im\phi} \\ &\times \sum_{n=-\infty}^{+\infty} H_n^{A, \tau, \eta}[f_n](b\rho) e^{in\phi} \sum_{k=-\infty}^{+\infty} i^k J_k(r\rho) e^{ik\theta} e^{-ik\phi} \rho d\rho d\phi \\ &= \frac{1}{2\pi} \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} i^{-m} J_m(\rho r_0) e^{-im\theta_0} \sum_{n=-\infty}^{+\infty} H_n^{A, \tau, \eta}[f_n](b\rho) \\ &\times \sum_{k=-\infty}^{+\infty} i^k J_k(r\rho) e^{ik\theta} \rho d\rho \int_0^{2\pi} e^{im\phi + in\phi - ik\phi} d\phi. \end{aligned} \quad (62)$$

Since

$$\int_0^{2\pi} e^{im\phi + in\phi - ik\phi} d\phi = 2\pi \delta_{m+n-k, 0} = \begin{cases} 2\pi, & m+n-k=0 \\ 0, & \text{otherwise,} \end{cases} \quad (63)$$

where $\delta_{\cdot, \cdot}$ denotes the Kronecker delta operator.

Hence, (62) can be written as

$$\int_0^{+\infty} \sum_{m=-\infty}^{+\infty} i^n J_m(\rho r_0) e^{-im\theta_0} \sum_{k=-\infty}^{+\infty} H_{k-m}^{A, \tau, \eta}[f_{k-m}](b\rho) J_k(r\rho) e^{ik\theta} \rho d\rho. \quad (64)$$

We make the following symbols

$$\tilde{z}(r, \theta) = e^{i \left[\frac{a}{2b} r^2 + \frac{r|\tau|}{b} \sin(\theta + \varphi_1) \right]} z(r, \theta), \quad (65)$$

$$\tilde{g}(r_0, \theta_0) = e^{i \left[\frac{a}{2b} r_0^2 + \frac{r_0|\tau|}{b} \sin(\theta_0 + \varphi_1) \right]} g(r_0, \theta_0), \quad (66)$$

and there is a Fourier series expansion

$$\tilde{g}(r_0, \theta_0) = \sum_{n=-\infty}^{+\infty} \tilde{g}_n(r_0) e^{in\theta_0}, \quad (67)$$

where $\tilde{g}_n(r_0) = w_2 e^{i \frac{a}{2b} r_0^2} g_n(r_0)$, w_2 is given by (17), and $g_n(r_0)$ is the n th term in the Fourier series for the function $g(r_0, \theta_0)$.

Using (59), (62) and (67), we get

$$\begin{aligned} \tilde{z}(r, \theta) &= \frac{1}{2\pi} \int_0^{+\infty} \int_0^{2\pi} \sum_{n=-\infty}^{+\infty} \tilde{g}_n(r_0) \sum_{m=-\infty}^{+\infty} i^n J_m(\rho r_0) r_0 d r_0 \\ &\times \sum_{k=-\infty}^{+\infty} H_{k-m}^{A, \tau, \eta}[f_{k-m}](b\rho) J_k(r\rho) e^{ik\theta} \int_0^{2\pi} e^{in\theta_0 - im\theta_0} d\theta_0 \rho d\rho. \end{aligned} \quad (68)$$

Letting $m = n$, and by (33), we obtain

$$\begin{aligned}
\tilde{z}(r, \theta) &= \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} i^n \int_0^{+\infty} \tilde{g}_m(r_0) J_m(\rho r_0) r_0 dr_0 \\
&\times \sum_{k=-\infty}^{+\infty} H_{k-m}^{A, \tau, \eta} [f_{k-m}](b\rho) J_k(r\rho) e^{ik\theta} \rho d\rho \\
&= \sum_{k=-\infty}^{+\infty} \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} i^n H_m[\tilde{g}_m(r_0)](\rho) \\
&\times H_{k-m}^{A, \tau, \eta} [f_{k-m}](b\rho) J_k(r\rho) e^{ik\theta} \rho d\rho.
\end{aligned} \tag{69}$$

Equation (69) is a Fourier series expansion; then, the k th term is

$$\tilde{z}_k(r) = \int_0^{+\infty} \sum_{m=-\infty}^{+\infty} i^n H_m[\tilde{g}_m(r_0)](\rho) H_{k-m}^{A, \tau, \eta} [f_{k-m}](b\rho) J_k(r\rho) \rho d\rho, \tag{70}$$

where $\tilde{z}_k(r) = w_2 e^{i\frac{d}{2b}r^2} z_k(r)$, w_2 is given by (17), and $z_k(r)$ is the k th term in the Fourier series for the function $z(r, \theta)$.

Applying the inverse HT formula, (70) becomes

$$H_k[\tilde{z}_k](\rho) = \sum_{m=-\infty}^{+\infty} i^n H_m[\tilde{g}_m](\rho) H_{k-m}^{A, \tau, \eta} [f_{k-m}](b\rho). \tag{71}$$

According to the relationship between the OLCHT and the HT, we obtain

$$\begin{aligned}
H_k^{A, \tau, \eta} [z_k](b\rho) &= \frac{i^k w_1 \ell e^{im(\varphi_1 - \varphi_2)}}{b} e^{i\frac{d}{2b}\rho^2} H_k[\tilde{z}_k](\rho), \\
H_m^{A, \tau, \eta} [g_m](b\rho) &= \frac{i^m w_1 \ell e^{im(\varphi_1 - \varphi_2)}}{b} e^{i\frac{d}{2b}\rho^2} H_m[\tilde{g}_m](\rho),
\end{aligned} \tag{72}$$

where φ_1 and φ_2 are given by (9), ℓ is given by (13), and w_1 is given by (17).

Hence, by (71), we have

$$H_k^{A, \tau, \eta} [z_k](\rho) = \sum_{m=-\infty}^{+\infty} H_m^{A, \tau, \eta} [g_m](\rho) H_{k-m}^{A, \tau, \eta} [f_{k-m}](\rho), \tag{73}$$

that is, a convolution relation

$$Z_k^{A, \tau, \eta}(\rho) = \sum_{m=-\infty}^{+\infty} G_m^{A, \tau, \eta}(\rho) F_{k-m}^{A, \tau, \eta}(\rho). \tag{74}$$

According to (49), the convolution of two sets of Fourier coefficients is equivalent to the multiplication of functions [32]. Therefore, it can be seen from (74) that there is a multiplicative relationship

$$Z^{A, \tau, \eta}(\rho, \phi) = F^{A, \tau, \eta}(\rho, \phi) G^{A, \tau, \eta}(\rho, \phi), \tag{75}$$

where $Z^{A, \tau, \eta}(\rho, \phi)$, $F^{A, \tau, \eta}(\rho, \phi)$ and $G^{A, \tau, \eta}(\rho, \phi)$ denote the 2D OLCT in polar coordinates of the functions $z(r, \theta)$, $f(r, \theta)$ and $g(r, \theta)$, respectively. $\square$

Remark 8. When $A = (a, b; c, d)$, $\tau = \mathbf{0}$, and $\eta = \mathbf{0}$, where a, b, c, d are real numbers satisfying $ad - bc = 1$, Theorem 3 is equivalent to Theorem 1 of [25]. It is a further extension of the existing results, and it is a perfection and supplement to the system of the LCT convolution theory.

Convolution theorem leads to the correlation theorem since the correlation of two functions is no greater than the convolution of one of the functions with inverse axis transformation and complex conjugation [3]. On the basis of Theorem 3, the correlation property of the 2D OLCT in polar coordinates is as follows:

Remark 9. The correlation operation $\star^{A, \tau, \eta}$ for two functions $f(r, \theta)$ and $g(r, \theta)$ in the OLCT domain can be expressed naturally as

$$(f \star^{A, \tau, \eta} g)(r, \theta) = O_L^{A^{-1}, \xi, \lambda} \left[F^{A, \tau, \eta}(\rho, \phi) \overline{G^{A, \tau, \eta}(\rho, \phi)} \right](r, \theta), \tag{76}$$

where $O_L^{A^{-1}, \xi, \lambda}$ denotes the inverse 2D OLCT operator, the bar denotes complex conjugation, $F^{A, \tau, \eta}(\rho, \phi)$ and $G^{A, \tau, \eta}(\rho, \phi)$ denote the 2D OLCT in polar coordinates of $f(r, \theta)$ and $g(r, \theta)$.

Similar to convolution, according to the expression form of (76), correlation operator and correlation theorem can be obtained. Research on more correlation property is our next work.

5. Applications of the 2D OLCT in polar coordinates

The 2D OLCT in polar coordinates is an effective processing tool in the fields of signal processing, medical imaging and photoacoustics. Next, we present some applications to illustrate the importance of the 2D OLCT in polar coordinates.

5.1. Applications in computed tomography (CT)

In this subsection, we discuss a specific application of the proposed 2D OLCT in the field of computed tomography (CT) as a typical example. It is well known that the direct Fourier technique to reconstruct CT is based on the projection theorem [26,33]. We let the signal $f(x, y)$ be space limited to W , and we let its FT in polar coordinates $F(\rho, \phi)$ be angularly bandlimited to K ; then, $F(\rho, \phi)$ can be reconstructed from its polar samples via [26,33]

$$F(\rho, \phi) = \sum_{n=-\infty}^{+\infty} \sum_{l=0}^{N-1} F\left(\frac{n}{W}, \frac{2\pi l}{N}\right) \text{sinc}\left[\frac{W(\rho - n)}{W}\right] \sigma\left(\phi - \frac{2\pi l}{N}\right), \quad (77)$$

where N is assumed even, $n \geq 0$, and

$$\sigma = \frac{\sin\left[\frac{1}{2}(N-1)\phi\right]}{N \sin\left(\frac{1}{2}\phi\right)}.$$

It can be seen from [26,27] that in practical applications, not all $f(r, \theta)$ functions are bandlimited. If the signal is not bandlimited in the FT domain, the direct Fourier technique does not work optimally in CT. Since conventional nonbandlimited signals in the FT domain may be bandlimited signals in the OLCT domain, we can obtain new insight by using the proposed method in this paper in the field of CT.

Based on the direct Fourier technique, we next discuss the reconstruction formulation of the 2D OLCT $F^{A,\tau,\eta}(\rho, \phi)$ in the CT field. Let $f(r, \theta)$ be angularly bandlimited to the highest frequency $\frac{K}{2\pi}$, and it is Ω -bandlimited in the OLCT domain if

$$F^{A,\tau,\eta}(\rho, \phi) = 0, \quad \rho \geq \Omega.$$

A relation exists between the 2D OLCT and the 2D FT

$$F^{A,\tau,\eta}(\rho, \phi) = \frac{e^{i\frac{d|\tau|^2}{b}}}{b} e^{i\left[\frac{d}{2b}\rho^2 - \frac{\rho|d\tau - b\eta|}{b} \sin(\phi + \varphi_2)\right]} F[\tilde{f}]\left(\frac{\rho}{b}, \phi\right), \quad (78)$$

where $\tilde{f}(r, \theta) = e^{i\left[\frac{a}{2b}r^2 + \frac{r|\tau|}{b} \sin(\theta + \varphi_1)\right]} f(r, \theta)$, φ_1 and φ_2 are given by (9). According to related theories of the n th-order Bessel function of the first kind, through (77) and (78), we derive the reconstruction formula for the 2D OLCT $F^{A,\tau,\eta}(\rho, \phi)$ of $f(r, \theta)$ at the normalized zeros α_{nj} and at the uniformly spaced points $\frac{2\pi l}{N}$ as follows:

$$\begin{aligned} F^{A,\tau,\eta}(\rho, \phi) &= (-1)^v \varsigma e^{i\frac{d}{2b}\rho^2} \sum_{l=0}^{N-1} \sum_{m=-\infty}^{+\infty} e^{-i\frac{a}{2b}\alpha_{vj}^2} J_m\left(\frac{|d\tau - b\eta|\rho}{b}\right) J_m^2\left(\frac{|\tau|\Omega}{b}\right) \\ &\quad \times \sum_{j=1}^{\infty} J_m\left(\frac{|\tau|\alpha_{vj}}{b}\right) \vartheta_{vj}(\rho) F^{A,\tau,\eta}\left(\alpha_{vj}, \frac{2\pi l}{N}\right) \sigma\left(\phi - \frac{2\pi l}{N}\right), \end{aligned} \quad (79)$$

where $\alpha_{nj} = \frac{bz_{nj}}{\Omega}$, z_{nj} is the j th zero of $J_n(z)$,

$$\varsigma = e^{i\left(\frac{d|\tau|^2 - a|b\eta - d\tau|^2}{b}\right)},$$

and

$$\vartheta_{nj}(r) = \frac{2b[\mu_2 + \alpha_{nj}] J_n\left(\frac{\Omega r}{b}\right)}{\Omega J_{n+1}(z_{nj}) [\alpha_{nj}^2 - r^2 + 2\mu_2(\alpha_{nj} - r)]}.$$

By changing σ , (79) can be used with N odd numbers, but in CT, where (79) has its main application, N is usually even.

We now outline how the OLCT in polar coordinates is used in the CT field. If $F^{A,\tau,\eta}(\rho, \phi)$ denotes the 2D OLCT of $f(x, y)$ in polar coordinates, then the projection theorem states that

$$P_\phi(\rho) = F^{A,\tau,\eta}(\rho, \phi), \quad \rho \geq 0, \quad 0 < \phi \leq 2\pi. \quad (80)$$

Equation (80) states that the OLCT of a projection is a central cross section of the OLCT of the image. In theory, if $F^{A,\tau,\eta}(\rho, \phi)$ were known everywhere, it would be a simple matter to convert this OLCT from polar to Cartesian coordinates; then, we can reconstruct the object function by OLCT inversion (6). The main steps are as follows:

- 1) Based on the idea of the central slice theorem, the projection value $P_\phi(\rho)$ under each projection angle is obtained, and the 1D OLCT $P_\phi(v)$ is solved.

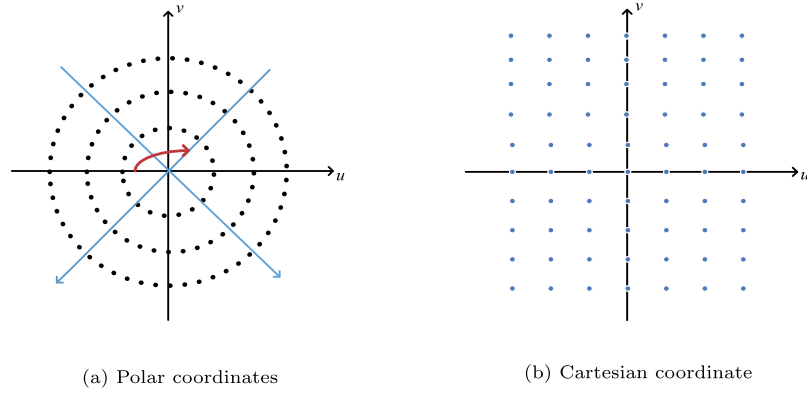


Fig. 1. Rasterization. The values corresponding to the coordinates of the black dots on the left are known, and the values corresponding to the coordinates of the blue dots on the right are obtained by interpolation. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

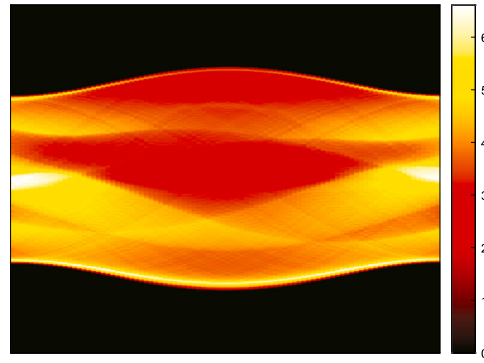


Fig. 2. Projected distribution map.

- 2) From the result of 1), the polar coordinate representation of the 2D OLCT of the image is obtained, that is, $F^{A,\tau,\eta}(\rho, \phi)$ is obtained on a uniform polar lattice by using (80).
- 3) The Cartesian coordinate $F^{A,\tau,\eta}(x, y)$ is obtained from the polar coordinate $F^{A,\tau,\eta}(\rho, \phi)$. The process of solving this problem is using two-dimensional interpolation, which is called “rasterization”. As shown in Fig. 1.
- 4) According to the result of 3), $f(x, y)$ is obtained by the inverse 2D OLCT.

In the above steps, the key step is to use (79) to interpolate $F^{A,\tau,\eta}(\rho, \phi)$ from a polar lattice to a Cartesian lattice. Let $N = 2K + 2$ and evaluate $F^{A,\tau,\eta}\left(\alpha_{vj}, \frac{2\pi l}{N}\right)$ at the appropriate Cartesian points $\{u_l, v_m, l, m \in \mathbb{N}\}$, where

$$\begin{aligned} \rho_{lm} &= \sqrt{u_l^2 + v_m^2} \geq 0, \\ \phi_{lm} &= \cos^{-1}\left(\frac{u_l}{\rho_{lm}}\right). \end{aligned} \quad (81)$$

Thus, $f(x, y)$ is obtained by the inverse 2D OLCT. Based on the above discussion, the practical application of the 2D OLCT on CT in polar coordinates is meaningful.

Taking the Shepp-Logan head model (S-L model) in the MATLAB image processing toolbox as an example, the simulation results are used to illustrate the feasibility of the proposed OLCT method. The phantom function in MATLAB is selected, the original image size is $256 * 256$, and the schematic structure of the S-L model is shown in Fig. 3 (a). The original image is scanned with parallel beams by the Radon transformation, and the projection angle is $0 \sim 179^\circ$ to obtain projection data. The projection distribution is shown in Fig. 2. During the reconstruction process, we select different 2D OLCT parameter matrices A , τ and η ($\det(A) = 1$) for the OLCT reconstruction. The reconstruction results are shown in Fig. 3. The simulation results show that the proposed method can reconstruct clear CT images well, and has certain feasibility in the CT field.

Besides visual effects, in order to evaluate the quality of the reconstructed images, three evaluation indicators are computed, i.e. the structural similarity index (SSIM), mean squared error (MSE) and peak signal to noise ratio (PSNR). The SSIM, MSE and PSNR are computed by MATLAB functions *ssim*, *immse* and *psnr* respectively. Larger SSIM means better reconstructed image quality and small MSE means higher similarities of the results and the original images. The higher the PSNR value, the less distortion. The SSIM, MSE and PSNR are compared for reconstruction under different OLCT parameter matrices in Table 1. From Table 1, the proposed method in this paper has a good effect of reconstructing images, and new insights can be obtained in the CT field, which provides more possibilities for the theoretical basis of the CT field.

Compared with the traditional FT, the proposed transform method in this paper has six free parameters and has more possibilities in sampling, time-frequency analysis, discrete algorithms and so on. It is possible that the nonbandlimited signal in the traditional FT domain is a bandlimited signal in the OLCT domain at a certain angle, so we can obtain new insight by using the proposed method in this paper

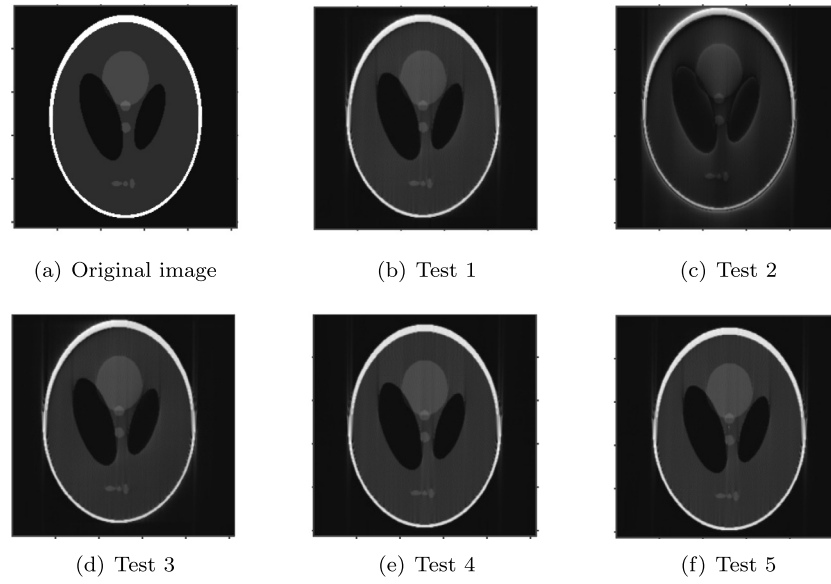


Fig. 3. Reconstruction results under different OLCT parameter matrices. (a) S-L model original image; (b) Test 1: $A = (0, 5/12; -12/5, 1/(3 \times 10^7))$, $\tau = (0, 0)$, $\eta = (1/55, 1/55)$; (c) Test 2: $A = (1/10^7, 3; -1/3, 0)$, $\tau = (0, 0)$, $\eta = (3/49, 3/49)$; (d) Test 3: $A = (0, 7/26; -26/7, 1/10^7)$, $\tau = (0, 0)$, $\eta = (1/50, 1/50)$; (e) Test 4: $A = (1/(2 \times 10^8), 3; -1/3, 0)$, $\tau = (1/(3 \times 10^7), 1/(3 \times 10^7))$, $\eta = (0, 0)$; (f) Test 5: $A = (0, -2/5; 5/2, 1/10^8)$, $\tau = (1/10^5, 1/10^5)$, $\eta = (1/89, 1/89)$.

Table 1

Comparison of SSIM, MSE and PSNR for CT reconstruction.

OLCT parameters	SSIM	MSE	PSNR
Test 1	0.3361	0.0552	12.5786
Test 2	0.3117	0.0548	12.6122
Test 3	0.3157	0.0546	12.6253
Test 4	0.4350	0.0552	12.5824
Test 5	0.4072	0.0549	12.6079

in sampling, time-frequency analysis and discrete algorithms. In addition to the reconstruction formula of the 2D OLCT $F^{A,\tau,\eta}(\rho, \phi)$ in the CT field, the proposed 2D OLCT in this paper has many novel properties and characteristics that urgently need to be studied, such as sampling theory, time-frequency analysis, and discrete fast algorithms.

5.2. Other potential applications

The first potential application of our research is sampling. It is well known that the signals in the field of color images based on traditional FT in polar coordinates are generally bandlimited signals. The most famous work in this area is Stark's work [34] that a function with limited bandwidth in the FT domain and limited highest frequency can be recovered perfectly from its samples at infinite normalized zeros of Bessel functions in radius and at finite uniform points in azimuth. However, it can be seen from [26,27] that in practical applications, not all $f(r, \theta)$ functions are bandlimited, so our results can be used to deal with nonbandlimited functions in the LCT domain or linear canonical Hankel transform (LCHT) domain. Based on the classical Stark's interpolation formulas, the sampling theorems of the bandlimited functions $f(r, \theta)$ in the OLCT and OLCHT domains can avoid the errors caused by the above methods. Thus, the sampling theorems for the 2D OLCT of the angular periodic bandlimited function in a polar coordinate system play an important role in many imaging fields, such as data compression, image digitization and encoding [35], which have broad application prospects.

There are also potential applications in the field of photoacoustics. Because the 2D OLCT has two more parameters than the LCT, it can achieve more functions than the LCT; that is, it has more application potential in optical information processing. The frequency domain in Fourier optics is also a special case in the transform domain of continuous distributions in the OLCT optics. Every property and every application of the Fourier transform is a special case of the OLCT. Therefore, the OLCT can be used to generalize and improve the FT in every field where the use of the FT and frequency domain concepts is well developed; hence, there is a broad area for the development of the OLCT optics.

6. Conclusions

According to the definition of the OLCT in the 2D Cartesian coordinate system, we obtain its new expression in polar coordinates and derives the mathematical formula of the OLCHT. Then, using the relationship between the 2D OLCT and the OLCHT, the spatial shift theorem and convolution theorem of the 2D OLCT are proved. Finally, some applications of the 2D OLCT in polar coordinates are given. These contents are new achievements in the field of the 2D OLCT in polar coordinates. In future work, we will consider the other related application implementation issues of the 2D OLCT by polar coordinates in optics and signal processing.

CRediT authorship contribution statement

Hui Zhao: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft. **Bing-Zhao Li:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- [1] R. Tao, B.Z. Li, Y. Wang, G.K. Aggrey, On sampling of bandlimited signals associated with the linear canonical transform, *IEEE Trans. Signal Process.* 56 (11) (2008) 5454–5464.
- [2] J. Shi, X.P. Liu, N.T. Zhang, Generalized convolution and product theorems associated with linear canonical transform, *Signal Image Video Process.* 8 (5) (2014) 967–974.
- [3] D.Y. Wei, Q.W. Ran, Y.M. Li, A convolution and correlation theorem for linear canonical transform and its application, *Circuits Syst. Signal Process.* 31 (2012) 301–312.
- [4] J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan, *Linear Canonical Transforms: Theory and Applications*, Springer, New York, 2016.
- [5] Y.N. Sun, B.Z. Li, Sliding discrete linear canonical transform, *IEEE Trans. Signal Process.* 66 (17) (2018) 4553–4563.
- [6] Q. Xiang, K.Y. Qin, Convolution, correlation, and sampling theorems for the offset linear canonical transform, *Signal Image Video Process.* 8 (3) (2014) 433–442.
- [7] D.Y. Wei, Y.M. Li, Convolution and multichannel sampling for the offset linear canonical transform and their applications, *IEEE Trans. Signal Process.* 67 (23) (2019) 6009–6024.
- [8] S.C. Pei, J.J. Ding, Eigenfunctions of the offset Fourier, fractional Fourier and linear canonical transforms, *J. Opt. Soc. Am. A* 20 (3) (2003) 522–532.
- [9] S.C. Pei, J.J. Ding, Eigenfunctions of Fourier and fractional Fourier transforms with complex offsets and parameters, *IEEE Trans. Circuits Syst. I* 54 (7) (2007) 1599–1611.
- [10] Z.W. Li, W.B. Gao, B.Z. Li, The solvability of a class of convolution equations associated with 2D FRFT, *Mathematics* 8 (11) (2020) 1–12.
- [11] L.B. Almeida, The fractional Fourier-transform and time-frequency representations, *IEEE Trans. Signal Process.* 42 (11) (1994) 3084–3091.
- [12] S.C. Pei, J.J. Ding, Relations between fractional operations and time-frequency distributions, and their applications, *IEEE Trans. Signal Process.* 49 (8) (2001) 1638–1655.
- [13] D.F.V. James, G.S. Agarwal, The generalized Fresnel transform and its application to optics, *Opt. Commun.* 126 (4–6) (1996) 207–212.
- [14] M. Moshinsky, C. Quesne, Linear canonical transformations and their unitary representations, *J. Math. Phys.* 12 (8) (1971) 1772–1783.
- [15] K.K. Sharma, S.D. Joshi, Signal separation using linear canonical and fractional Fourier transforms, *Opt. Commun.* 265 (2) (2006) 454–460.
- [16] S. Xu, L. Feng, Y. Chai, Y. He, Analysis of A-stationary random signals in the linear canonical transform domain, *Signal Process.* 146 (2018) 126–132.
- [17] A. Stern, Sampling of compact signals in offset linear canonical domains, *Signal Image Video Process.* 1 (4) (2007) 359–367.
- [18] Q. Xiang, K.Y. Qin, Q.Z. Huang, Multichannel sampling of signals band-limited in offset linear canonical transform domains, *Circuits Syst. Signal Process.* 32 (5) (2013) 2385–2406.
- [19] A. Elgargati, R. Daher, Uncertainty principles for the offset linear canonical transform, *J. Pseudo-Differ. Oper. Appl.* 12 (3) (2021) 1–8.
- [20] H.Y. Huo, W.C. Sun, L. Xiao, Uncertainty principles associated with the offset linear canonical transform, *Math. Methods Appl. Sci.* 42 (2) (2019) 466–474.
- [21] Y. Xu, M.H. Xu, L.H.V. Wang, Exact frequency-domain reconstruction for thermoacoustic tomography-II: cylindrical geometry, *IEEE Trans. Med. Imaging* 21 (7) (2002) 829–833.
- [22] B. Rigaud, A. Simon, M. Gobeli, et al., Statistical shape model to generate a planning library for cervical adaptive radiotherapy, *IEEE Trans. Med. Imaging* 38 (2) (2019) 406–416.
- [23] J.D. Suever, G.J. Wehner, L.Y. Jing, et al., Right ventricular strain, torsion, and dyssynchrony in healthy subjects using 3D spiral cine DENSE magnetic resonance imaging, *IEEE Trans. Med. Imaging* 36 (5) (2017) 1076–1085.
- [24] G. Chirikjian, A. Kyatkin, *Engineering Applications of Noncommutative Harmonic Analysis: With Emphasis on Rotation and Motion Groups*, Academic, New York, NY, USA, 2001.
- [25] Z.C. Zhang, Convolution theorem for two-dimensional LCT of angularly periodic functions in polar coordinates, *IEEE Signal Process. Lett.* 26 (8) (2019) 1142–1146.
- [26] A.I. Zayed, Sampling of signals bandlimited to a Disc in the linear canonical transform, *IEEE Signal Process. Lett.* 25 (12) (2018) 1765–1769.
- [27] Z.C. Zhang, A. Sun, Z.Y. Liang, et al., Sampling theorems for bandlimited function in the two-dimensional LCT and the LCHT domains, *Digit. Signal Process.* 114 (1) (2021) 103053, 1–8.
- [28] I. Gradshteyn, I. Ryzhik, *Tables of Integrals, Series, and Products*, Academic, New York, NY, USA, 1965.
- [29] J.V. Cornacchio, R.P. Soni, On a relation between two-dimensional Fourier integrals and series of Hankel transforms, *J. Res. Natl. Bur. Stand. B, Math. Math. Phys.* 69B (3) (1965) 173–174.
- [30] N. Baddour, Two-dimensional Fourier transforms in polar coordinates, *Adv. Imaging Electron Phys.* 165 (2011) 1–45.
- [31] N. Baddour, Operational and convolution properties of two-dimensional Fourier transforms in polar coordinates, *J. Opt. Soc. Am. A* 26 (8) (2009) 1767–1777.
- [32] A. Oppenheim, R. Schaffer, *Discrete-Time Signal Processing*, Prentice-Hall, Englewood Cliffs, NJ, USA, 1989.
- [33] R.J. Marks II (Ed.), *Advanced Topics in Shannon Sampling and Interpolation Theory*, Springer-Verlag, New York, USA, 1993, pp. 188–191.
- [34] H. Stark, Sampling theorems in polar coordinates, *J. Opt. Soc. Am.* 69 (11) (1979) 1519–1525.
- [35] J.W. Goodman, *Introduction to Fourier Optics*, McGraw Hill, 1968.

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